JUSTICE OHENE AMOFA

Noguchi Memorial Institute for Medical Research
P.O. Box LG 581, University of Ghana, Legon, Accra, Ghana
https://github.com/iamamofa | Justiceoheneamofa@gmail.com | https://www.linkedin.com/in/joamofa/
https://iamamofa.github.io

PROFESSIONAL SUMMARY

A highly motivated and results-driven early-career scientist with a multidisciplinary background in bioinformatics, computational biology, and biotechnology, with experience supporting infectious disease research and genomic surveillance. Skilled in data analysis, molecular modeling, and development of reproducible scientific workflows, with a strong interest in translating computational insights into practical applications in vaccine research and biomanufacturing. Experienced in working within interdisciplinary teams, contributing to pathogen analysis, and supporting training and capacity-building initiatives in public health genomics across Africa. Demonstrates a solid foundation in laboratory data systems, standard operating procedures, and scientific best practices relevant to regulated environments.

My goal is to utilize my research and programming expertise to address significant challenges in public health and beyond.

KEY SKILLS

Programming: R, Python, Rust, PHP, Bash Scripting, Nfcore, Grovy, C#, C++, MATLAB, OCTAVE
Machine Learning / Data Science: Scikit-learn, PyTorch, KERAS, TensorFlow, Theano, ML Algorithms
Software Engineering Frameworks: Flutter, MERN stack, Django, Flask
DataBases Management: MySQL, PostgreSQL, MongoDB
Cloud/Containerization: Docker, Singularity, HPC
Application Programming Interface (API): Rest API, Fast API
Tools and Frameworks: Nextflow, Snakemake, GISAID, SeekDeep Pipeline, Bacpage Bactmap, Cholgen, DESeq2, Qiime2, Harmonization, Mamba and Conda Packages.
High Performance Computing: HPC Clusters, Distributed Systems, Nodes, WSU Grid, LENGAU, Tesla V100 GPU
Document Creation & Referencing: Microsoft Office Suite, Mendeley, Zotero
Modelling & Docking Software: Gaussian, VMD, ChimeraX, Schrödinger Suite, Alphafold Server, PyMOL, Cobafold, AutoDock Vina, GOLD, Spartan, Avogadro, ChemDraw *Chemical, AutoDock Tools, Discovery Studio Visualizer, Chimera X*

EDUCATION

BSc | ***Computer Science***
University of Ghana, Legon
School of Mathematical Sciences - College of basic and Applied Sciences

BSc | ***Health Science***
University of the People, Pasadena - California, United States of America (USA)
School of Sciences - Public Health Science

ASc | ***Health Science (*associate degree)**
University of the People, Pasadena - California, United States of America

RESEARCH & WORK EXPERIENCE

Public Health Alliance for Genomic Epidemiology (PHA4GE)
Training Coordinator & Co-Lead Bioinformatics Author *– Cholera Guidance Document (Driver Project Award)*
Supervisors: Dr. Tracey Calvert-Joshua, Prof. Alan Christoffels
2026 – Present

- ✦ Selected for a competitive international Driver Project to co-lead development of a cholera bioinformatics guidance framework for global public health genomics.
- ✦ Designing standardized workflows for *Vibrio cholerae* genomic analysis, including QC, assembly, AMR profiling, phylogeny, and data interpretation.
- ✦ Collaborating with international experts through biweekly working group meetings to produce a community-driven, consensus-based guidance document.
- ✦ Contributing to global standards for reproducible pathogen genomics pipelines and capacity building in low-resource settings.
- ✦ Leading the manuscript development and technical writing of a community-driven, consensus-based bioinformatics guidance document for publication and global dissemination.

Noguchi Memorial Institute for Medical Research
Genomics Research Assistant & Bioinformatics (*60 hours/week)*
Supervisors: Dr. Bright Adu and Prof. Dorothy Yeboah-Manu, *2023 – Present*

✦ Conducted DNA and RNA quantification and quality assessment using NanoDrop spectrophotometry, Qubit fluorometry, and agarose gel electrophoresis to ensure suitability for downstream sequencing applications.

✦ Prepared sequencing libraries for next-generation sequencing (NGS), including DNA fragmentation, end repair, adapter ligation, PCR amplification, and library quality control following standardized laboratory protocols.

✦ Engages in the analysis of bacterial pathogens, including cholera, tuberculosis, Klebsiella, and various infectious viruses, contributing to critical public health research and surveillance.

✦ Applies structural bioinformatics and computational biology approaches to study pathogen biology, including *Mycobacterium tuberculosis, Vibrio cholerae,* and other clinically relevant microorganisms.

✦ Utilizes protein modeling, molecular visualization, and docking tools (e.g., AlphaFold, PyMOL, ChimeraX, AutoDock) to investigate biomolecular structures and interactions relevant to drug discovery and vaccine target identification.

✦ Contributes to genomics-informed pathogen surveillance, integrating sequencing data with structural and functional analysis to support public health and translational research.

✦ Supports development and optimization of bioinformatics pipelines for genomic and amplicon data, aligning workflows with reproducibility and standardization principles applicable to regulated environments.

✦ Assists in the development of bioinformatics analysis pipelines for whole genome and amplicon sequencing data, improving the efficiency of data processing and analysis.

✦ Supervise and train interns and students on Bioinformatics workflows, techniques, protocols and facilitate bioinformatics and genomics training programs organized by external organizations, such as the Africa CDC, hosted at our institute and tailored for various African countries towards capacity building.

Noguchi Memorial Institute for Medical Research *Onsite (Accra) 2023 - present*
Teaching Assistantship – One Health *(40 hours/week)* *Supervisors:* Dr. Prince Asare and Prof. Dorothy Yeboah-Manu,

✦ Assisted in the teaching and practical delivery of postgraduate courses in Pathogen Genomics and Bioinformatics, Research Methods and Biostatistics, Molecular Cell Biology, Pathogen–Host Immunology, Dynamics of Antimicrobial Resistance, and related One Health curricula for MPhil and PhD students.

- ✦ Facilitated laboratory demonstrations and hands-on computational training in microbial genomics, next-generation sequencing (NGS) data analysis, genome assembly, variant analysis, phylogenetics, and pathogen surveillance using Linux-based bioinformatics workflows.
- ✦ Designed, developed, and maintained research software applications and bioinformatics tools to support teaching, laboratory activities, and collaborative research projects; gathered user requirements and produced clean, efficient, and well-documented code.
- ✦ Assisted students and researchers in the development and optimization of reproducible bioinformatics pipelines for whole-genome sequencing, amplicon sequencing, and metagenomic analyses using workflow management and version control systems.
- ✦ Supported course assessments, practical examinations, and research projects by providing technical guidance on genomics, computational biology, and pathogen genomic epidemiology.

USQRD & Omnitech Solutions Limited *Remote, London Area, United Kingdom*
Large Language Models (LLM) Engineer *(40 hours/week)* *Supervisor* – Mr. Daniel Usyvat

- ✦ Developed and optimized large language models (LLMs) for various applications, enhancing the performance and accuracy of data-driven solutions.
- ✦ Collaborated with cross-functional teams to integrate LLM capabilities into existing systems, facilitating improved user interactions and data processing workflows and conducted extensive testing and evaluation of LLM outputs to ensure reliability and adherence to project specifications.
- ✦ Engaged in data preprocessing and feature engineering to enhance model training, resulting in significant improvements in model performance and Assisted in the documentation of processes and methodologies, contributing to knowledge sharing and best practices within the team.

Theoretical and Computational Chemistry - Department of Chemistry – University of Ghana
Accra, Ghana
Research Intern (*40 hours/week*)

- ✦ Provided chemical and mechanistic insight into a series of selected organic reactions using DFT
- ✦ Applied Density Functional Theory (DFT) and quantum chemical methods to investigate reaction mechanisms, energy profiles, and molecular stability of organic and biologically relevant compounds.
- ✦ Performed molecular modeling and electronic structure analysis to understand structure–function relationships, with relevance to drug design and biomolecular interactions.
- ✦ Utilized computational tools including Gaussian 09, Spartan 14, Avogadro, ChemDraw, and CYLview for geometry optimization, visualization, and simulation of chemical systems.
- ✦ Conducted in silico analysis of reaction pathways, identifying transition states, intermediates, and thermodynamic/kinetic properties of chemical reactions.
- ✦ Gained hands-on experience in computational approaches used in medicinal chemistry and early-stage drug discovery, including ligand optimization and molecular interaction analysis.
- ✦ Interpreted computational results to provide mechanistic insights supporting experimental chemistry workflows.
- ✦ Developed strong foundations in computational chemistry, molecular design, and chemical informatics, applicable to pharmaceutical and biotechnological research.

Genomics and Molecular Biology (Wet Laboratory) Experience

Noguchi Memorial Institute for Medical Research (NMIMR), University of Ghana – Accra, Ghana
***Genomics & Bioinformatics Core Facility- Research Intern* | 202 – Present**

- ✦ Performed high-quality genomic DNA and RNA extraction from bacterial isolates and clinical specimens using manual and commercial extraction protocols.
- ✦ Conducted DNA and RNA quantification and quality assessment using NanoDrop spectrophotometry, Qubit fluorometry, and agarose gel electrophoresis to ensure suitability for downstream sequencing applications.

- Prepared sequencing libraries for next-generation sequencing (NGS), including DNA fragmentation, end repair, adapter ligation, PCR amplification, and library quality control following standardized laboratory protocols.
- Assisted in whole-genome sequencing (WGS) and targeted amplicon sequencing workflows for bacterial and viral pathogens, including Mycobacterium tuberculosis, Vibrio cholerae, Klebsiella pneumoniae, and other clinically relevant microorganisms.
- Supported operation and maintenance of sequencing workflows, including sample preparation, sequencing run setup, and post-run quality assessment in collaboration with the Genomics Core Facility.
- Performed PCR, multiplex PCR, and amplicon generation for pathogen detection, molecular characterization, and sequencing applications.

Trestle Academy, 4th IR *Onsite (Accra) , Remotely (Switzerland)*
Data Engineer *(40 hours/week)* *Supervisor* – Mr. Dereke Djede / Wonders Aggor

- Design, develop and maintain software Application with collaborations to gather requirements and also writes clean, efficient and well documented code and also troubleshoot, debug and resolve software defects.
- Implement statistical models for data analysis and identify trends, patterns and anomalies in datasets. Utilizes visualization tools for effective storytelling and also employ version control systems to manage code and analytical scripts.

SELECTED HONOURS AND AWARDS

- Full Travel Grant for Africa CDC - NGS Academy Bioinformatics Training On Pathogen Surveillance 2024 , Cape Town, South Africa)
- Government of Ghana merit-based Scholarship for higher education (Date of award: September 2015. Duration: November **2015 – 2018**)
- Full Scholarship, University of the People (BSc Health Science), awarded full scholarship amounting to $7,400 for course assessment fees.
- Full travel Sponsorship and scholarship for Vibriowatch and Cholera Genomics Course (CIDRZ) – Lusaka, Zambia.
- Full Sponsorship award for Biostruct -Africa Workshop Empowering Africa-based Scientist through structural biology knowledge- ILRI, Nairobi, Kenya.
- Full Sponsorship (Oral Presentation – Harnessing Functional Genomics in Cancer Research Opportunities for Diagnosis and Treatment – Windhoek, Namibia.
- Full Travel Award and Sponsorship (PHA4GE Conference 2025) (Workshop and Abstract Presentation) – Data for Action: Overcoming Challenges and Seizing Opportunities in Public Health Genomics, Cape Town, South Africa.
- Full Travel Sponsorship 1st Metagenomic Virus Discovery Workshop (CERI – Stellenbosch, South Africa)
- Travel Grant for AI & AMR Symposium, African Center of Excellence in Bioinformatics & Data Intensive Sciences, Uganda.
- Best Oral Presentation at the, AI & AMR Symposium, African Center of Excellence in Bioinformatics & Data Intensive Sciences, Uganda.
- Travel Award scholarship Pathogen Data Network (PDN) CUREs Workshop , Cold Spring Harbor Laboratory (CSHL), DNA Learning Center, CSHL, 1 Bungtown Road, Cold Spring Harbor, New York, USA, March 2026
- Travel Grant for WCB 2026 Congress, from 8 - 10 July 2026 at the Birchwood Hotel & Conference Centre, South Africa Johannesburg.
- Full travel Sponsorship and scholarship for 6th Big Data Africa School by South African Radio Astronomy Observatory (SARAO), Cape Town, South Africa.
- Full travel Sponsorship and scholarship for AI, Uncertainty & Simulation (Bayesian Methods and Neural Inference) Graduate Research and Workshop , Mauritius.

- ✦ Full travel Sponsorship and scholarship for Biotech Innovation Bootcamp 2026, Translational Research & Innovation Bootcamp, Ashesi University, Accra – Ghana.
- ✦ Travel Grant (Early Career Researcher Grant) for 21st International Congress for Infectious (ICID Conference), Madrid, Spain.

FACILITATING EXPERIENCE

- ✦ CDC-ASLM Bacterial Genomic Surveillance with a Focus on Antimicrobial Resistance – *Facilitator*
- ✦ H3ABioNeT & Wellcome Connecting Science Genomic Sequencing Bioinformatics (GSB) Africa – *NMIMR Classroom Facilitator*.
- ✦ AI for Clinicians: Leveraging AI tools for Clinical Practice, Biomedical Research and Grant Writing – *Facilitator* (Noguchi Memorial institute for medical Research)

SELECTED PRESENTATIONS

- ✦ **Oral Presentation**: *"Genomic Approaches to Bacterial Antimicrobial Resistance Surveillance"* – Wellcome Connecting Sciences Workshop on Bacterial Pathogen Genomics, Johannesburg, South Africa. Organized by Wellcome Connecting Sciences and NICD
- ✦ **Bacteriology Journal Club Presentation**: *"Analyzing Bacteriological Surveillance Data Using Genomics Approaches"* – Monthly Bacteriology Department Journal Club, Noguchi Memorial Institute for Medical Research.
- ✦ **Immunology Journal Club Presentation**: Presented on *"Immune Response Trends in Pathogen Genomics Surveillance"* at the Noguchi Memorial Institute for Medical Research, Immunology Department, ARL Training Lab.
- ✦ **Genomics and Bioinformatics Core Facility Journal Club:** Presented on "Public Health Alliance for Genomic Epidemiology, Advancing Public Health through Genomic Epidemiology" at the Bioinformatics Lab, Noguchi Memorial Institute for Medical Research.

EXTRA CURRICUULAR ACTIVITIES AND VOLUNTEERING SERVICES

- ✦ **Leader and Coordinator,** *Cholera Genomics Data Analysis Working Group*
  A working great initiative that was organized by Prof. Nicolas Thomson and Dr. Jolynne Mokaya (Sanger Institue , UK) as a result of the Cholera Training to come together with a great innovative mind to curate pipelines, data and assist ongoing countries who are endemic and provide analysis reports to policy decision makers about the endemic in their country. https://choleragenomicsdataanalysis.github.io/
- ✦ **Nextflow Ambassador,** *Seqera Labs , NFCore and Nextflow Company*

A working initiative to designed to empower passionate individuals within the Nextflow community to play a more active role in fostering collaboration, knowledge sharing, and engagement. I get the opportunity to amplify the reach and impact of Nextflow by promoting its adoption, organizing community events, sharing your expertise through presentations and tutorials, and assisting fellow community members with their questions and challenges. https://www.nextflow.io/our_ambassadors.html

- ✦ **Software Engineer Educator,** *Trestle Academy*
  One of the educators where I shared my background and stories of success. Regarding this platform, I I use it judiciously to provide best tips and practices that students can take in order to improve their way of life to become a better software Engineer and go for a correct career path.
  https://www.trestleacademyghana.org/alumni
- ✦ **NGS Code Clube Facilitator***, Genomics Core Facility Unit Lab – Noguchi Mem. Institute for Medical Research* (NMIMR) A code club set up in the Institute to bring each one to the Tech World of Genomics and machine learning.
- ✦ **Volunteer,** Smile on Faces Foundation

a foundation set up to raise money Together with a team, used to educate the people in the community, support orphanages and street children with technical skills like, cooking, beads making, saponification, and formal education. etc

- **Co-Lead Author** – Cholera Bioinformatics Guidance, PHA4GE , Contributor – Wastewater Surveillance Standards (SARS-CoV-2 / Influenza)

PROFESSIONAL MEMBERSHIPS

Open Bioinformatics Foundation (OBF)| American Chemical Society (ACS) | SAGESA - Society for African Genomic Surveillance of Antimicrobial Resistance | Bioinformatics & Computational Biology (B&CB) | PHA4GE- Bioinformatics Pipelines and Visualization Working Group | NGS Academy, Africa PGI | Google Flutter App Developers | Microsoft Learn Student Ambassadors | American Society for Microbiology (ASM) | NGS code Club|Cholgen Group| NfCore Ambassador

CONFERENCES, WORKSOPS, SAMPLE PROJECTS AND REFEREES

Below Embedded | Reference, conferences, workshops and mini projects.

SAMPLE PROJECTS AND UNDERTAKINGS

- **Flux Inventory System:** This system is designed for institutes to efficiently manage their inventory. It keeps track of stock levels, ensuring that supplies are never depleted unexpectedly, and aids in maintaining optimal inventory for operational continuity**.** I continue to manage and fix bugs in relation to the system. **App Url (**https://flux-inc.onrender.com/)
- **TB Detection App – Genomic Sequencing Bioinformatics (GSB) Africa – NMIMR Classroom Facilitator:** This application facilitates the detection of tuberculosis through advanced genomic sequencing techniques. As part of the GSB Africa program at the Noguchi Memorial Institute for Medical Research (NMIMR), this role involves teaching and guiding students in using bioinformatics to analyze genomic data for TB detection**. App Url (**https://iamamofa-tb-detection-tb-detectionappapp-xlqsyp.streamlit.app/)
- **PANGenS Web Training Portal and PANGenS Web Site–** Built and management of PANGenS
- Website. The Web Training portal is an online Learning management site to management the courses to be taken freely by consortium members. And also manages the consortium website. **App Urls (**https://genofluxcore.com/ **,** https://pangens.org/ **)**
- **GoodReads Data Pipeline:** Built a data pipeline where **d**ata is captured in real time from the goodreads API using the Goodreads Python wrapper (View usage - Fetch Data Module). The data collected from the goodreads API is stored on local disk and is timely moved to the Landing Bucket on AWS S3. ETL jobs are written in spark and scheduled in airflow to run every 10 minutes. **(**https://github.com/iamamofa/Final_project_Justice_goodreads_etl_pipeline**)**
- **NMIMR Cholegen Analyzer –**NMIMR, the Cholegen Analyzer, built a pipeline to undertake and automate analysis of cholera pathogen. Its an updated version that was previously started by another institute in Ghana.It is currently under construction to be converted to a web app. The pipeline can currently be run only locally. **App Url (**https://github.com/seqafrica/bacpage_nmimr**)**

PUBLICATIONS –PUBLISHED & UNPUBLISHED RESEARCH ARTICLES AND PAPERS)

- **Amofa, J. O.**, George, J., Okumu, N. A., Ohene, M., Terefe, E. M., Tsomene, S. L., et al. (2026). *BioStruct-Africa's scalable framework for AlphaFold-enabled research training and sustainable workforce development in Africa.* Communications Biology, 9, Article 321.
- Charlix Rix , **Amofa, J.**, et al. (2026). *Africa's emerging role in precision oncology: Translational insights from the Harnessing Functional Genomics in Cancer Research Conference, Windhoek, 23–26 September 2025.* **BMC Proceedings, 20**, 30. https://doi.org/10.1186/s12919-026-00387-z
- Smith, E. A., Ahmadi Jeshvaghane, M., Pilz, D. A., Shitindo, M., Costa da Silva, J., **Amofa, J. O.,** et al. (2026).
Community-led standards for global wastewater-based infectious disease surveillance. *PLOS Global Public Health*, 6(4), e0006266.
- Mokaya J, Ibrahim RA, **Amofa JO**, et al. The future of cholera genomics through the lens of clinical, public health, and research practitioners in Africa. The Lancet Microbe. Published online October 9, 2026. Article 101528. In press, corrected proof.
- Kameni, G. N., Mbamyah, E. L., Ndedi, E. D. F. M., Kamgang, I. D. N., Ikomey, G. M., Kunkeng, C. Y., Mesembe, M. T., Elimbi, N. Y., Kameni, M. N., Fokou, J. B. T., **Amofa, J. O.**, Toyim, A. N., & Nyegue, M. A. (2025). *Carbapenem-Resistant Pseudomonas aeruginosa: A Multicenter Study on Prevalence, Distribution, and Risk Factors across Six Health Facilities in Yaoundé, Cameroon.* International Journal of Medical Science and Clinical Research Studies, 5(3), 439–451.
- Smith, E. A., Ahmadi Jeshvaghane, M., Pilz, D. A., Shitindo, M., Costa da Silva, J., et al., **Amofa, J. O.**, et al. *Community-Led Standards for Global Wastewater-Based Infectious Disease Surveillance.* Preprint. Public Health Alliance for Genomic Epidemiology (PHA4GE).
- Lebika Bangoup MM, Ndoye FF, Kameni GN, Noubissi TA, Nganou IDK, Fai DK**, Ohene Justice Amofa**, et al. Phenotypic Antibiotic Resistance Profiles of Clinical Escherichia coli and Klebsiella pneumoniae Isolates from Cameroon: Prevalence of ESBL, Carbapenemase, and Multidrug Resistance. Access Microbiology. September 2026. Preprint; under review.
- Kameni GN, Mbamyah LE, Moni NEDF, Nganou IDK, Noubissi TA, Nana CO, Yengo CK, **Justice Ohene Amofa** et al. Molecular Characterization of Carbapenemase-Encoding Genes in Carbapenem-Resistant Pseudomonas aeruginosa Isolated from Clinical Samples in Yaound√©, Cameroon. Access Microbiology. August 2026. Preprint; under review.
- Nganou Kamgang ID, Kameni GN, Moni NEDF, Boyomo Onana, Nana CO, Noubissi TA, Njima NJCH, **Ohene Justice Amofa,** et al. Investigating Conjugation-Mediated Spread of ESBLs and Carbapenemases Genes in Clinical Pseudomonas aeruginosa, Escherichia coli and Klebsiella pneumoniae in Yaound√©, Cameroon. Access Microbiology. August 2026. Preprint; under review.
- **Amofa, J. Ohene,** M.Najah, Agyenkwa-Mawuli . K, Asare. P, Adu. B, Otchere. I.D, Yeboah-Manu D. (2024). Using AI and machine learning to predict TB infections based on patient demographics, risk factors, and clinical presentations, optimizing early detection and reducing diagnostic delays.
- **Amofa, J. Ohene**, M.Najah, Agyenkwa-Mawuli . K, Gyampo .K, Asare. P, Adu. B, Otchere. I.D,Yeboah-Manu D. (2024). Integrating digital health solutions to address cyberbulling's impact on youth mental health and support chronic condition management in adolescents.

- Kwasi Atweri Akowah, Evangeline Obodai, Grace Arezie Kyire, Roland Elikem Anlonya, **Justice Ohene Amofa**, Evelyn Yayra Bonney, Ivy Asantewaa Asante, Nana Afia Asante Ntim, Joseph Humphery Kofi Bonney and John Kofi Odoom*. Nosocomial infection of SARS-CoV-2 among laboratory staff in a Biomedical Research Institution

WORKSHOP AND CONFERENCES

- **SeqShere+ Training: TB Genomics Data Analysis -**Noguchi Memorial Institute for Medical Research – PANGenS (Pan-Africa Network for Genomics Surveillance of Poverty Related Disease and Emerging Pathogens)
- **Malaysia-Japan bilateral Workshop on Bioinformatics and Bioresource -** National Institute of Genetics, Mishima, Japan
- **NGS Academy Bioinformatics Training on Pathogen Surveillance -** South African National Bioinformatics Institute (SANBI), University of the Western Cape, Cape Town. South Africa
- **Genomics for Antimicrobial Resistance Surveillance and One Health**, Wellcome Connecting Sciences & NICD, Johannesburg, South Africa
- **Genomics workshop on SeekDeep pipeline for Paired end Illumina Amplicon Targeted Sequencing Analysis**, Noguchi Memorial Institute for Medical Research, Accra, Ghana
- **AI2050 AI/ML Tools in Drug Discovery Workshop**, H3D Foundation and Ersilia – Accra, Ghana
- **DSI-Africa: First Human Genome Repeat Expansion Analysis Virtual Workshop**
- Data Science for Health Discovery and Innovation in Africa, Consortium, South Africa **Genomic Sequence Bioinformatics (Certificate)**
- **H3ABioNet** – Pan African Bioinformatics Network for H3Africa **(NMIMR Classroom)**
- **Vibriowatch and Cholera Genomics Course, Centre for Infectious Disease Research (CIDRZ) -** Zambia
- **Ghana genomics workshop on Molecular Inversion Probe Design and Population Genetics Analysis**
  - Noguchi Memorial Institute for Medical Research, Accra, Ghana
- **DS-ZOOFOOD Workshop on Infectious Disease Modelling,** University of Ghana (Department of Medical Microbiology- University of Ghana Medical School)
- **Introduction to Bioinformatics in Bacterial Genomics** (By Cambridge University and NMIMR), Accra Ghana
- **12 Months Software Engineering Training (Certificate)** - ALX Africa (online)
- **Other Major Crash Courses on Python, Flutter, Data Science, etc (Certificate)** Coursera, Udemy, DataCamp and Zindi Africa

REFEREES

Prof. Dorothy Yeboah Manu
Director / Professor / Principal Investigator
Noguchi Memorial Institute for Medical Research
Email: dyeboah-manu@noguchi.ug.edu.gh
Tel: +233208123882

Dr. Bright Adu
Principal Investigator / Senior Research Fellow /
Coordinator - **Genomics and Bioinformatics Core Facility and Africa CDC Regional Sequencing Lab, NMIMR**
Noguchi Memorial Institute for Medical Research
Email: badu@noguchi.ug.edu.gh
Tel: +233268446661

Dr. Kwabena Owusu Danquah
Senior Research Fellow / Principal Investigator
Department of Clinical Pathology
Noguchi Memorial Institute for Medical Research (Deputy Head of Department)
Email: kdanquah@noguchi.ug.edu.gh

Dr. Jolynne Mokaya
Senior Research Fellow / Engagement Lead
Sanger Institute (UK), Africa CDC, PHA4GE
Email: Jm61@sanger.ac.uk
Tel: +44 7468955097

Tel: +233543913833

Prof Heather Moore
Dean / Provost for Health Science
University of the People
Email: Heather.moore.uoopeople.edu

Prof. Kofi Sarpong Adu-Manu
Senior Lecturer / Associate Professor
Department of Computer Science
University of Ghana
College of Basic and Applied Science
Email: ksadu-manu@ug.edu.gh
Tel: +233244602374

Mr. Robert Wonders Aggor
Founder & Director of Innovation and Entrepreneurship
Trestle Academy
Email: wonders@trestleacademyghana.org
Tel: +1 (226) 784-5489